Workshop Sheet - Week 1

Reading and writing a Lean proof. InfoView displays your hypotheses and current goal. Place the cursor inside a proof to see its state; replace each `sorry` with a proof.

`intro h`	Assume the premise of an implication.
`exact h`	Submit `h` as a proof of the current goal; Lean checks that it matches.
`rw [h]`	Rewrite using an equality; add `at hsq` to change a hypothesis.
`rfl`	Prove an equality whose two sides are equal by definition.
`refine ⟨k, ?_⟩`	Supply a witness and leave its required property as a new goal.
`cases h`	Split a hypothesis into its possible cases.
`rcases h with ...`	Unpack a witness or split a disjunction into cases.
`have h : P := by ...`	State and prove an intermediate claim.
`by_contra h`	Assume the goal is false and seek a contradiction.
`simp`	Simplify the goal using standard rewrite rules.
`simpa using h`	Simplify the goal and `h`'s statement, then finish with `h`.
`norm_num`	Normalize numbers: simplify and prove numerical facts.
`linarith`	Combine linear equations and inequalities ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$).
`nlinarith`	Deduce consequences of polynomial equalities and inequalities.

Indentation groups proof steps; · starts a separate proof branch.

1. *Tactic practice.*
   (a) Prove that any proposition P implies itself.
   (b) Given $k \in \mathbb{N}$, prove that there exists $m \in \mathbb{N}$ with $m = k$.
   (c) For integers a, b, deduce $a + 1 = b + 1$ from $a = b$.
   (d) For $m \in \mathbb{N}$, prove $(m + 1) - m = 1$.
   (e) For $n \in \mathbb{N}$, if $n = 2$ or $n = 3$, prove $n^2 \leq 9$.
   (f) For $x, y \in \mathbb{R}$ with $x + y \leq 7$ and $3 \leq y$, prove $x \leq 4$.
   (g) For $x \in \mathbb{R}$, prove $x^2 + 1 > 0$.
   (h) Verify that $\frac{3}{4} + \frac{1}{6} = \frac{11}{12}$ in $\mathbb{Q}$.

2. *Infinitely many primes.*
   (a) If $n > 0$ and a prime p divides n, prove $2 \leq p \leq n$.
   (b) Prove that no prime divides two consecutive natural numbers.
   (c) Prove that 2 is the only even prime.
   Hint. `Nat.dvd_prime hp` says $d \mid p$ if and only if $d = 1$ or $d = p$.
   (d) Prove that $n! + 1$ has a prime divisor.
   Finding facts. Use `loogle.lean-lang.org` to look up `Nat.factorial_pos` and `Nat.exists_prime_and_dvd`. The latter needs $n! + 1 \neq 1$. Hover over a lemma or use `#check` outside a proof to inspect its assumptions and conclusion.
   (e) Suppose p is prime, $p \leq n$ and $p \mid n! + 1$. Derive a contradiction.
   Hint. Search Loogle for `"dvd", "factorial"` (`dvd` means *divides*).
   Use `Nat.dvd_factorial` with `hp.pos` to get $p \mid n!$, then `Nat.dvd_sub` and `simpa` to get $p \mid 1$. Finish with `hp.not_dvd_one`.
   (f) Formalize the statement and proof of the following theorem.
   Theorem. *There are infinitely many prime numbers.*

3. *The irrationality of $\sqrt{2}$.*
   A rational square root would give coprime integers n, d, with $d > 0$ and $n^2 = 2d^2$. The next steps force both numerator and denominator to be even.
   (a) For $k \in \mathbb{Z}$, prove $2 \mid 2k$ by supplying a witness for divisibility.
   (b) For $n \in \mathbb{Z}$, deduce $2 \mid n$ from $2 \mid n^2$.
   Hint. Inspect `Int.prime_two.dvd_mul`; use `.mp hsq` and split the cases with `rcases`.
   (c) Substitute $n = 2k$ into $n^2 = 2d^2$ and deduce $d^2 = 2k^2$.
   Hint. Rewrite the hypothesis, then use `nlinarith`.
   (d) Show that coprime natural numbers cannot both be even.
   Hint. `Nat.Coprime.of_dvd` would make 2 coprime to itself; `norm_num` gives a contradiction.
   (e) Formalize the statement and proof of the following theorem.
   Theorem. *The number $\sqrt{2}$ is irrational.*